DSC5001: Statistical Machine Learning I

Tutorial 7: Kernel and Gaussian Process Regression

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Task 1: Kernel Regression and a Distant Point

Four input-response pairs are observed:

$$x = (1, 2, 3, 6), \qquad y = (2, 4, 7, 30).$$

Predict at $x_0 = 2.5$, using the triangular kernel

$$K(u) = \max(1 - |u|, 0), \qquad w_i = K\left(\frac{x_i - x_0}{h}\right).$$

(a) For $h = 2$, compute the raw weights w_i and normalized weights $\alpha_i = w_i / \sum_j w_j$.

(b) Find the Nadaraya–Watson prediction $\widehat{f}_2(2.5) = \sum_i \alpha_i y_i$ and effective sample size $n_{\text{eff}} = 1/\sum_i \alpha_i^2$.

(c) Repeat the raw weights, normalized weights, and prediction for $h = 4$.

(d) If the last response increases from 30 to 50, how much does the prediction change for each bandwidth?

Solution: Task 1(a) – Weights at $h = 2$

For $x_0 = 2.5$ and $h = 2$, the scaled distances are

$$\frac{x_i - x_0}{h} = \left(-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{7}{4}\right).$$

Using $K(u) = \max(1 - |u|, 0)$, the raw weights are

$$\boxed{w = \left(\frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 0\right).}$$

Their total is $7/4$, giving

$$\boxed{\alpha = \left(\frac{1}{7}, \frac{3}{7}, \frac{3}{7}, 0\right).}$$

Solution: Task 1(b) – Prediction and Information

Using $y = (2, 4, 7, 30)$ and the normalized weights from (a),

$$\widehat{f}_2(2.5) = \frac{1}{7}(2) + \frac{3}{7}(4) + \frac{3}{7}(7) + 0(30).$$

Therefore,

$$\boxed{\widehat{f}_2(2.5) = 5}.$$

For effective sample size,

$$n_{\text{eff}} = \frac{1}{(1/7)^2 + (3/7)^2 + (3/7)^2}.$$

Hence,

$$\boxed{n_{\text{eff}} = \frac{49}{19} \approx 2.579}.$$

Solution: Task 1(c) – Wider Bandwidth

At $x_0 = 2.5$ with $h = 4$,

$$\frac{x_i - x_0}{4} = \left(-\frac{3}{8}, -\frac{1}{8}, \frac{1}{8}, \frac{7}{8}\right).$$

The triangular kernel gives

$$\boxed{w = \left(\frac{5}{8}, \frac{7}{8}, \frac{7}{8}, \frac{1}{8}\right)}.$$

Since $\sum_i w_i = 5/2$,

$$\boxed{\alpha = \left(\frac{1}{4}, \frac{7}{20}, \frac{7}{20}, \frac{1}{20}\right)}.$$

Then

$$\widehat{f}_4(2.5) = \frac{1}{4}(2) + \frac{7}{20}(4) + \frac{7}{20}(7) + \frac{1}{20}(30).$$

Thus $\boxed{\widehat{f}_4(2.5) = \frac{117}{20} = 5.85}$.

Solution: Task 1(d) – Response Change

Only the last response changes, from 30 to 50:

$$\Delta y_4 = 50 - 30 = 20.$$

At $h = 2$, its normalized weight is zero:

$$\boxed{\Delta \widehat{f}_2 = 0 \times 20 = 0}.$$

At $h = 4$, its normalized weight is $1/20$:

$$\boxed{\Delta \widehat{f}_4 = \frac{1}{20}(20) = 1}.$$

$$\boxed{\widehat{f}_2^{\text{new}}(2.5) = 5, \qquad \widehat{f}_4^{\text{new}}(2.5) = 6.85}.$$

Task 2: Local Linear Fitting

Near the left edge of an input range, we observe

$$x = (1, 2, 4), \qquad y = (4, 5, 9), \qquad x_0 = 1.$$

For the triangular kernel $K(u) = \max(1 - |u|, 0)$ and $h = 4$, the raw weights are

$$w = \left(1, \frac{3}{4}, \frac{1}{4}\right).$$

Fit $y_i \approx a + b(x_i - x_0)$. Let $u_i = x_i - x_0$, let $\boldsymbol{X}_0$ have rows $[1, u_i]$, and let $\boldsymbol{W}_0 = \mathrm{diag}(w_i)$.

(a) Compute the local constant prediction at $x_0 = 1$.

(b) Construct $\boldsymbol{X}_0$ and $\boldsymbol{W}_0$.

(c) Compute $\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{X}_0$ and $\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{y}$, where $\boldsymbol{y} = (4, 5, 9)^\top$. Solve for the local line and its prediction at x_0.

(d) Compare the weighted training errors $Q = \sum_i w_i (y_i - \widehat{y}_i)^2$ of the local constant and local linear fits.

Solution: Task 2(a) – Local Constant

The weights sum to

$$1 + \frac{3}{4} + \frac{1}{4} = 2.$$

The weighted response sum is

$$1(4) + \frac{3}{4}(5) + \frac{1}{4}(9) = 10.$$

Therefore, the local constant prediction is

$$\boxed{\widehat{f}_{\mathrm{LC}}(1) = \frac{10}{2} = 5}.$$

Solution: Task 2(b) – Local Matrices

Since $x_0 = 1$, the centered inputs are

$$u_i = x_i - x_0 = (0, 1, 3).$$

The design and weight matrices are

$$\boxed{\boldsymbol{X}_0 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}} \qquad \boxed{\boldsymbol{W}_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}}.$$

The first column fits the local intercept, and the second fits the local slope.

Solution: Task 2(c) – Normal Equations

From $u = (0, 1, 3)$ and $w = (1, 3/4, 1/4)$,

$$\sum_i w_i = 2, \quad \sum_i w_i u_i = \frac{3}{2}, \quad \sum_i w_i u_i^2 = 3.$$

Therefore,

$$\boxed{\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{X}_0 = \begin{bmatrix} 2 & 3/2 \\ 3/2 & 3 \end{bmatrix}}.$$

Also,

$$\sum_i w_i y_i = 10, \qquad \sum_i w_i u_i y_i = \frac{21}{2},$$

so

$$\boxed{\boldsymbol{X}_0^\top \boldsymbol{W}_0 \boldsymbol{y} = \begin{bmatrix} 10 \\ 21/2 \end{bmatrix}}.$$

Solution: Task 2(c) – Solve the Local Line

Using the matrices from part (c), solve

$$\begin{bmatrix} 2 & 3/2 \\ 3/2 & 3 \end{bmatrix} \begin{bmatrix} \widehat{a} \\ \widehat{b} \end{bmatrix} = \begin{bmatrix} 10 \\ 21/2 \end{bmatrix}.$$

The coefficients are

$$\boxed{\widehat{a} = \frac{19}{5} = 3.8}, \qquad \boxed{\widehat{b} = \frac{8}{5} = 1.6}.$$

At $x = x_0 = 1$, the centered input is zero, so

$$\boxed{\widehat{f}_{\mathrm{LL}}(1) = \widehat{a} = 3.8}.$$

Solution: Task 2(d) – Local Training Errors

For the local constant $\widehat{f}_{\mathrm{LC}}(1) = 5$,

$$Q_{\mathrm{LC}} = 1(4-5)^2 + \frac{3}{4}(5-5)^2 + \frac{1}{4}(9-5)^2.$$

Thus $\boxed{Q_{\mathrm{LC}} = 5}$.
For the local line $\widehat{y}_i = 3.8 + 1.6(x_i - 1)$,

$$\widehat{y} = (3.8, 5.4, 8.6), \qquad y - \widehat{y} = (0.2, -0.4, 0.4).$$

The weighted error is

$$Q_{\mathrm{LL}} = 1(0.2)^2 + \frac{3}{4}(-0.4)^2 + \frac{1}{4}(0.4)^2.$$

Therefore $\boxed{Q_{\mathrm{LL}} = 0.2}$.
A better training fit does not necessarily imply better test prediction.

Task 3: GP and Sensor Noise

A machine's response Y is its vibration anomaly relative to normal operation. One sensor records $y_1 = +3$ at speed $x_1 = 1000$ rpm. Predict the hidden anomaly at $x_* = 1100$ rpm.
Assume a **zero-mean GP**. The prior function variance at both speeds is 1, and their prior covariance is $c = k(x_1, x_*) = 0.6$.
Compare two possible noise levels for this same measured value:

$$\text{Sensor A: } \sigma_n^2 = \frac{1}{4}, \qquad \text{Sensor B: } \sigma_n^2 = 1.$$

Use

$$\mu_* = \frac{c}{1+\sigma_n^2}\, y_1, \qquad s_*^2 = 1 - \frac{c^2}{1+\sigma_n^2}.$$

(a) For each sensor, find the variance $C = 1 + \sigma_n^2$ of the observed response.

(b) Find the posterior mean μ_* for each sensor.

(c) Find the posterior function variance s_*^2 for each sensor. Which sensor provides more information?

(d) For Sensor A, calculate the variance of a future Sensor A measurement and approximate 95% intervals for the hidden function and that measurement, using mean ± 1.96 standard deviations.

Solution: Task 3(a) – Training Variance

The prior function variance at the observed speed is 1. Independent observation noise adds σ_n^2:

$$C = 1 + \sigma_n^2.$$

For Sensor A,

$$\boxed{C_A = 1 + \frac{1}{4} = \frac{5}{4} = 1.25}.$$

For Sensor B,

$$\boxed{C_B = 1 + 1 = 2}.$$

Sensor B gives a noisier observation of the same hidden function.

Solution: Task 3(b) – Posterior Means

The one-observation GP prediction is

$$\mu_* = \frac{c}{C}\, y_1, \qquad c = 0.6, \quad y_1 = 3.$$

Sensor A gives

$$\boxed{\mu_{*,A} = \frac{0.6}{1.25}(3) = 1.44}.$$

Sensor B gives

$$\boxed{\mu_{*,B} = \frac{0.6}{2}(3) = 0.90}.$$

Higher observation noise leads to a smaller update from the prior mean of zero.

Solution: Task 3(c) – Function Uncertainty

The remaining function variance is

$$s_*^2 = 1 - \frac{c^2}{C}, \qquad c^2 = 0.36.$$

For Sensor A,

$$\boxed{s_{*,A}^2 = 1 - \frac{0.36}{1.25} = 0.712}.$$

For Sensor B,

$$\boxed{s_{*,B}^2 = 1 - \frac{0.36}{2} = 0.820}.$$

Sensor A has lower noise and reduces uncertainty more strongly.

Solution: Task 3(d) – Two Intervals

For Sensor A, $\mu_* = 1.44$, $s_*^2 = 0.712$, and $\sigma_n^2 = 0.25$.
The next measurement contains fresh noise:

$$\boxed{\mathrm{Var}(y_* \mid y_1) = 0.712 + 0.25 = 0.962}.$$

An approximate 95% interval for the hidden function is

$$1.44 \pm 1.96\sqrt{0.712} \quad \Longrightarrow \quad \boxed{[-0.21,\ 3.09]}.$$

For the next measured response, the interval is

$$1.44 \pm 1.96\sqrt{0.962} \quad \Longrightarrow \quad \boxed{[-0.48,\ 3.36]}.$$

The measurement interval is wider because it includes fresh noise.

Task 4: GP Denoising at a Training Input

Two soil sensors at depths $x_1 = 10$ cm and $x_2 = 20$ cm record noisy moisture anomalies

$$\boldsymbol{y} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

We want the **hidden** value at $x_* = 10$ cm, where the first measurement was made. Assume a zero-mean GP with covariance and independent noise variance

$$\boldsymbol{K} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}, \qquad \sigma_n^2 = \frac{1}{4}.$$

Since $x_* = x_1$, $\boldsymbol{k}_* = (1, 1/2)^\top$ and $k_{**} = 1$. Define $\boldsymbol{C} = \boldsymbol{K} + \sigma_n^2 \boldsymbol{I}$.

(a) Calculate $\boldsymbol{C}$. Why does independent noise enter only its diagonal?

(b) Solve $\boldsymbol{C}\boldsymbol{a} = \boldsymbol{y}$ and $\boldsymbol{C}\boldsymbol{b} = \boldsymbol{k}_*$.

(c) Calculate $\mu_* = \boldsymbol{k}_*^\top \boldsymbol{a}$ and $s_*^2 = k_{**} - \boldsymbol{k}_*^\top \boldsymbol{b}$.

(d) Find the variance of a new measurement at 10 cm. Why can μ_* differ from the measured value $y_1 = 1$?

Solution: Task 4(a) – Noisy Covariance

The prior covariance of the hidden function is

$$\boldsymbol{K} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}.$$

Independent observation noise adds $1/4$ only to diagonal entries:

$$\boldsymbol{C} = \boldsymbol{K} + \frac{1}{4}\boldsymbol{I}.$$

Therefore,

$$\boxed{\boldsymbol{C} = \begin{bmatrix} 5/4 & 1/2 \\ 1/2 & 5/4 \end{bmatrix}}.$$

Cross-covariance does not change because the two noise terms are independent.

Solution: Task 4(b) – First Linear System

Solve $\boldsymbol{C}\boldsymbol{a} = \boldsymbol{y}$:

$$\begin{bmatrix} 5/4 & 1/2 \\ 1/2 & 5/4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Multiplying by four gives

$$5a_1 + 2a_2 = 4, \qquad 2a_1 + 5a_2 = 12.$$

The solution is

$$\boxed{\boldsymbol{a} = \begin{bmatrix} -4/21 \\ 52/21 \end{bmatrix}}.$$

Solution: Task 4(b) – Second Linear System

Now solve $\boldsymbol{C}\boldsymbol{b} = \boldsymbol{k}_*$:

$$\begin{bmatrix} 5/4 & 1/2 \\ 1/2 & 5/4 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}.$$

Multiplying by four gives

$$5b_1 + 2b_2 = 4, \qquad 2b_1 + 5b_2 = 2.$$

The solution is

$$\boxed{\boldsymbol{b} = \begin{bmatrix} 16/21 \\ 2/21 \end{bmatrix}}.$$

Solution: Task 4(c) – Hidden-Value Prediction

Recall $\boldsymbol{k}_* = (1, 1/2)^\top$ and $k_{**} = 1$.
The posterior mean is

$$\mu_* = \boldsymbol{k}_*^\top \boldsymbol{a} = -\frac{4}{21} + \frac{1}{2}\left(\frac{52}{21}\right)$$
$$= \boxed{\frac{22}{21} \approx 1.048}.$$

The posterior function variance is

$$s_*^2 = 1 - \boldsymbol{k}_*^\top \boldsymbol{b} = 1 - \left(\frac{16}{21} + \frac{1}{2}\left(\frac{2}{21}\right)\right)$$
$$= \boxed{\frac{4}{21} \approx 0.190}.$$

Solution: Task 4(d) – Noisy Measurement

For a new measurement at the same depth, add fresh noise variance:

$$\begin{aligned}\mathrm{Var}(y_* \mid \boldsymbol{y}) &= s_*^2 + \sigma_n^2 \\ &= \frac{4}{21} + \frac{1}{4}.\end{aligned}$$

Therefore,

$$\boxed{\mathrm{Var}(y_* \mid \boldsymbol{y}) = \frac{37}{84} \approx 0.440}.$$

The observed value $y_1 = 1$ contains noise. The GP combines both measurements rather than treating y_1 as the exact hidden function value.

Tutorial Summary

After this tutorial, you should be able to:

- explain how bandwidth controls sensitivity to distant observations;
- compare local constant and local linear training fits;
- quantify how observation noise affects a GP prediction;
- explain GP denoising at a training input.

Tutorial 7: local fitting, measurement noise, and uncertainty

Questions and discussion